



Fully coupled modeling for static and dynamic analysis of thermo-magneto-electro-elastic plates

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Received: 8 July 2025 / Accepted: 5 November 2025
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Abstract Magneto-electro-elastic (MEE) materials with integrated sensory and actuation functions have promising application potentials in the spacecraft and automotive engineering. The deformation of MEE structures in temperature varying environments leads to a thermo-magneto-electro-elastic coupled problem. However, most studies considered temperature changes as a thermal load. This unidirectional coupling misses thermal changes due to structural deformations. This study establishes a thermo-magneto-electro-elastic fully coupled finite element (FE) model for MEE plates based on the first-order shear deformation (FOSD) theory. The proposed eight-node

Serendipity plate/shell elements for MEE structures incorporate five displacement DOFs per node, and one electric, one magnetic and one temperature DOF per element. The present FE model is first analyzed for mesh convergence. Subsequently, the static and dynamic responses of MEE plates under mechanical, electric, and magnetic loads are investigated. The numerical results, including displacements and temperature fields, are compared with COMSOL simulations. The comparative results indicate that the model has good accuracy in predicting the impact of structural deformations on the temperature field.

Keywords MEE plates · First-order shear deformation theory · Multi-physics coupling · Finite element analysis

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1 Introduction

Thin-walled structures are essential in aeronautical and astronautical engineering for lightweight design, but traditional single-function materials cannot meet the demanding requirements of modern structural applications. Magneto-electro-elastic (MEE) composite materials with their multi-functional properties have emerged as one of the innovative research topics in the mechanical engineering. Suchtelen (1972) first developed MEE materials by using ferroelectric phases (BaTiO_3) and ferromagnetic phases (CoFe_2O_4). This kind of composite materials

integrate the properties of ferroelectric and ferromagnetic materials, demonstrating dual functionality through electric field induced polarization and magnetic field induced magnetization. Their unique coupling effects - piezoelectric, piezomagnetic and magnetoelectric - enable diverse applications in sensors and actuators, energy harvesters, sonar applications, magnetic field probes, and microwave resonators (Pouresmaeeli et al. 2013; Liu and Lyu 2020; Shishesaz et al. (2018); Semenov et al. (2006)). To fully harness these coupling effects, it is crucial to investigate the structural behavior of MEE structures under multi-physics conditions.

Existing studies on the mechanical behavior of MEE structures have employed various computational approaches, with the primary methodologies being semi-analytical solution, finite element (FE) analysis and analytical method. Huang et al. (2007) derived closed-form solutions for key mechanical quantities including axial force and displacement fields in MEE beams to investigate plane stress behavior. Zhang et al. (2022) constructed a multi-physics coupled FE model and performed static and dynamic analysis on FG-MEE structures, and the proposed FE model is of great significance in the accurate design of FG-MEE structures. Zhou et al. (2022) utilized the enriched finite element method to evaluate the vibrational performance of MEE structures, effectively smoothing stress discontinuities between elements. Zhang et al. (2019, 2020a) proposed a semi-analytical approach combining the scaled boundary finite element method and the accurate integration technique to analyze the static deflection and natural vibration properties of MEE plates.

In addressing the governing equations of mechanical problems, several researchers have proposed feasible and efficient solution methodologies. The high-order nonlocal operator method, the nonlocal strong formulations for various physical models, and the variational method proposed by Ren et al. (2020, 2023, 2025) demonstrate remarkable computational efficiency and accuracy, providing useful insights for extension to MEE structures. Rabczuk et al. (2019) proposed a novel nonlocal operator theory for solving governing equations, which completely avoids the use of shape functions and greatly simplifies the solution process. Samaniego et al. (2020) presented a deep energy method (DEM), incorporating deep neural networks, for solving various mechanical and

multiphysics coupling problems. Nguyen-Thanh et al. (2021) proposed a Parametric Deep Energy Method (P-DEM) based on physics-informed neural networks (PINNs) to solve strain gradient elasticity problems. Eshaghi et al. (2025) proposed the Variational Physics-Informed Neural Operator (VINO), which embeds the variational energy form of partial differential equations (PDEs) into the neural operator framework. PDEs are solved by minimizing the energy form of PDEs, and VINO exhibits outstanding performance in scenarios with scarce data and high resolution. Although recent variational and machine learning approaches offer efficient alternatives, the FE method remains a well-established technique with recognized accuracy in multiphysics problems and the advantage of systematic validation through commercial software packages.

MEE materials are often used in complex environments, the static and dynamic behaviors of MEE structures under different loadings have been widely investigated. Researchers have examined MEE structures under mechanical, electric, magnetic, thermal, and hygrothermal fields, focusing primarily on how external loadings influence deformation, vibration, and stability. For instance, investigations have addressed bending and transient responses under combined electric, magnetic and mechanical fields (Zhao et al. 2022); vibrations under explosive, thermal, electric, and magnetic conditions (Dat et al. 2022); static and dynamic characteristics under various thermal loadings (Zhao et al. 2024); mechanical responses in hygrothermal environments (Vinyas and Kattimani 2017a, 2018a); and transient behaviors under moving heat flux (Tarkashvand et al. 2025). While these works provide valuable insights, most emphasize external excitations, whereas studies on fully coupled multiphysics effects remain limited.

The unique capability of MEE materials to facilitate coupling interactions between mechanical, electric, and magnetic fields (Priya et al. 2007), resulting in field induced electric and magnetic potentials, has attracted some research attention. Sladek et al. (2013) calculated the potential changes of MEE thick plates under intensity loading. Zhang et al. (2022) examined FG-MEE structures subjected to coupled mechanical, magnetic and electric loads, quantifying the induced potentials. Vinyas and Kattimani (2017b, 2018b) investigated multilayered METE beams and plates under varying thermal conditions, analyzing their effects

on displacements, potentials and magnetic flux density. Zhou and Qu (2023) conducted a comprehensive analysis of MEE structures subjected to thermal and mechanical loads, characterizing both static deformation and transient response while determining the resulting displacement and potential fields.

Previous research has primarily concentrated on output electric and magnetic potentials. However, due to the thermal expansion coefficients of MEE materials, externally applied physical fields can induce structural deformation, which in turn leads to changes in the temperature field of the structure, resulting in temperature differences. Cui et al. (2019) proposed a novel thermo-mechanical coupling approach using thermal deformation and material viscosity-induced deformation heating to describe the interconversion of internal energy and mechanical work. However, their study primarily focused on energy variations caused by deformation heating, without detailed analysis of the temperature field variations. Furthermore, there are almost no studies on the effect of structural deformation on the temperature field. With regard to this, this study seeks to establish a fully coupled model of magneto-electro-thermo-elastic for MEE plates within the first-order shear deformation (FOSD) theory to fill the research gap. Based on the principle of virtual work and Hamilton's principle, the governing equations are systematically derived and subsequently solved through a self-coded FE program to numerically investigate both static and transient responses of MEE plates. Beyond analyzing deformation effects under multiphysics fields, this work explicitly explores the resulting temperature field variations. The theoretical framework and computational findings presented in this article are projected to provide a crucial reference for the design optimization of MEE structures under various physical fields.

2 Theoretical formulations

2.1 Constitutive relations

Given the assumption of small strains in plates and shells, the linear constitutive relations for MEE materials are given by (Tassi et al. 2022)

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon} - \mathbf{e}^T \mathbf{E} - \mathbf{q}^T \mathbf{H} - \boldsymbol{\lambda}^T \boldsymbol{\theta}, \quad (1)$$

$$\mathbf{D} = \mathbf{e}\boldsymbol{\varepsilon} + \boldsymbol{\chi}\mathbf{E} + \boldsymbol{\kappa}\mathbf{H} + \mathbf{p}^T \boldsymbol{\theta}, \quad (2)$$

$$\mathbf{B} = \mathbf{q}\boldsymbol{\varepsilon} + \boldsymbol{\kappa}\mathbf{E} + \boldsymbol{\gamma}\mathbf{H} + \boldsymbol{\tau}^T \boldsymbol{\theta}, \quad (3)$$

$$\mathbf{S} = \boldsymbol{\lambda}\boldsymbol{\varepsilon} + \mathbf{p}\mathbf{E} + \boldsymbol{\tau}\mathbf{H} + \mathbf{c}\boldsymbol{\theta}, \quad (4)$$

$$\boldsymbol{\lambda}^T = \mathbf{C}\boldsymbol{\alpha}, \quad (5)$$

$$\boldsymbol{\theta} = \mathbf{T} - \mathbf{T}_0. \quad (6)$$

Here, $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$, $\mathbf{D}$ and $\mathbf{E}$ denote the stress, strain, electric displacement and electric field vectors, respectively; $\mathbf{B}$, $\mathbf{H}$, $\mathbf{S}$ and $\boldsymbol{\theta}$ denote the magnetic induction, magnetic field, entropy and temperature difference vectors, respectively. Furthermore, $\mathbf{C}$, $\boldsymbol{\chi}$ and $\boldsymbol{\gamma}$ denote the elastic, dielectric, magnetostrictive constant matrices, respectively; $\mathbf{e}$, $\mathbf{q}$ and $\boldsymbol{\kappa}$ denote the piezoelectric, piezomagnetic and magnetoelectric coupling coefficient matrices, respectively; $\boldsymbol{\lambda}$, $\boldsymbol{\alpha}$, $\mathbf{p}$, $\boldsymbol{\tau}$ and $\mathbf{c}$ denote the thermal stress moduli, thermal expansion, pyroelectric coefficient, pyromagnetic coefficient and thermal capacity matrices, respectively.

Assuming that the plate and shell thickness is small relative to the in-plane dimensions, the normal stresses σ_{33} are negligible compared to the planar stresses. Accordingly, the comprehensive matrix representation corresponding to Eqs. 1–4 is relegated to Appendix A, with the main text focusing on the core derivation steps.

Under the assumption that all applied fields (electric, magnetic and thermal) are thickness-directional only, which leads to $\chi_{11} = \chi_{22} = 0$, $\kappa_{11} = \kappa_{22} = 0$, $\gamma_{11} = \gamma_{22} = 0$, $p_1 = p_2 = 0$, $\tau_1 = \tau_2 = 0$. In accordance with Maxwell's equations, the electric and magnetic fields are derived from their respective potentials through gradient operations, specifically

$$\mathbf{E} = -\nabla \boldsymbol{\phi} = \mathbf{B}_{\boldsymbol{\phi}} \boldsymbol{\phi} = \left\{ -\frac{\partial \phi_1}{\partial \Theta^1}, -\frac{\partial \phi_2}{\partial \Theta^2}, -\frac{\partial \phi_3}{\partial \Theta^3} \right\}, \quad (7)$$

$$\mathbf{H} = -\nabla \boldsymbol{\psi} = \mathbf{B}_{\boldsymbol{\psi}} \boldsymbol{\psi} = \left\{ -\frac{\partial \psi_1}{\partial \Theta^1}, -\frac{\partial \psi_2}{\partial \Theta^2}, -\frac{\partial \psi_3}{\partial \Theta^3} \right\}, \quad (8)$$

where ∇ is the gradient operator, $\boldsymbol{\phi}$ and $\boldsymbol{\psi}$ represent the electric and magnetic potential vectors, respectively; $\mathbf{B}_{\boldsymbol{\phi}}$ and $\mathbf{B}_{\boldsymbol{\psi}}$ denote the electric and magnetic field matrix, respectively.

Since the electric and magnetic fields act in the thickness direction, the in-plane components can be neglected. The electric and magnetic potentials are

postulated to exhibit linear variation along the thickness direction. Eqs. 7 and 8 can be simplified to

$$E_3 = -\frac{\phi_3}{h_E}, \quad (9)$$

$$H_3 = -\frac{\psi_3}{h_H}, \quad (10)$$

where h_E and h_H are the distances between electrodes and magnetic poles of the current layer. Moreover, ϕ_3 and ψ_3 represent the imposed electric magnetic potentials along the thickness direction, respectively.

Due to the continuous variation of temperature difference along the thickness direction, the structure is discretized into N layers for calculation. The distribution of the temperature field within the structure is represented by the temperature difference in each layer, as expressed below:

$$\theta = \mathbf{B}_T \theta = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_N \end{Bmatrix}, \quad (11)$$

where $\mathbf{B}_T$ is the temperature distribution matrix and θ_i denotes the temperature difference in the i -th layer.

2.2 Strain–displacement field

This paper adopts the FOSD theory, which states that straight lines perpendicular to the mid-plane of the structure stay straight after deformation, but may no longer be perpendicular to the mid-plane, and the structural thickness remains unchanged. Under this theory, the displacement field of any point within plate/shell structures can be characterized by five independent kinematic parameters, expressed as (Zhang et al. 2015)

$$v_\alpha = v_\alpha^0 + \Theta^3 v_\alpha^1, \quad (12)$$

$$v_3 = v_3^0, \quad (13)$$

where α takes 1 and 2, v_α^0 and v_3^0 represent the translational displacements, v_α^1 are the generalized rotational displacements. These five parameters can be seen as degrees of freedom ($u, v, w, \varphi_1, \varphi_2$) as presented in

Fig. 1. Eqs. 12 and 13 can be written in the following matrix form

$$\mathbf{u} = \begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \Theta^3 & 0 \\ 0 & 1 & 0 & 0 & \Theta^3 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{Bmatrix} 0 \\ v_1 \\ 0 \\ v_2 \\ 0 \\ v_3 \\ 1 \\ v_1 \\ 1 \\ v_2 \end{Bmatrix} = \mathbf{Z}_u \mathbf{v}. \quad (14)$$

Here, $\mathbf{v}$ denotes the generalized displacement vector.

Disregarding the compression of the structure along the thickness direction, the strain tensors including only the in-plane strains $\varepsilon_{\alpha\beta}$ and transverse shear strains $\varepsilon_{\alpha 3}$, which are obtained as (Zhang et al. 2020b)

$$\varepsilon_{\alpha\beta} = \varepsilon_{\alpha\beta}^0 + \Theta^3 \varepsilon_{\alpha\beta}^1 + (\Theta^3)^2 \varepsilon_{\alpha\beta}^2, \quad (15)$$

$$\varepsilon_{\alpha 3} = \varepsilon_{\alpha 3}^0 + \Theta^3 \varepsilon_{\alpha 3}^1, \quad (16)$$

where $\varepsilon_{11}^0, \varepsilon_{22}^0$ are the in-plane longitudinal strains, $\varepsilon_{12}^0, \varepsilon_{21}^0$ denote the in-plane shear strains, $\varepsilon_{13}^0, \varepsilon_{23}^0$ represent the transverse shear strains, $\varepsilon_{11}^1, \varepsilon_{22}^1$ denote the bending strains, $\varepsilon_{12}^1, \varepsilon_{21}^1$ are the torsional strains, and $\varepsilon_{\alpha\beta}^2, \varepsilon_{\alpha 3}^1$ represent corrections for the in-plane and shear strains, respectively.

Each strain component in Eqs. 15 and 16 can be expressed as

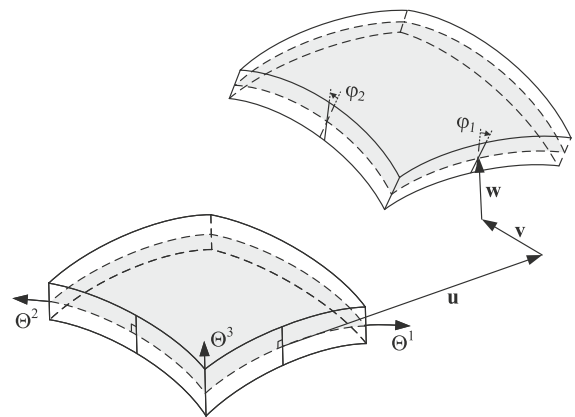


Fig. 1 Degrees of freedom

$$2\varepsilon_{\alpha\beta}^0 = \varphi_{\alpha\beta}^0 + \varphi_{\beta\alpha}^0, \quad (17)$$

$$2\varepsilon_{\alpha\beta}^1 = \varphi_{\alpha\beta}^0 - b_{\beta}^{\lambda} \varphi_{\lambda\alpha}^0 + \varphi_{\beta\alpha}^1 - b_{\alpha}^{\delta} \varphi_{\delta\beta}^0, \quad (18)$$

$$2\varepsilon_{\alpha\beta}^2 = -b_{\beta}^{\lambda} \varphi_{\lambda\alpha}^1 - b_{\alpha}^{\delta} \varphi_{\delta\beta}^1, \quad (19)$$

$$2\varepsilon_{\alpha 3}^0 = v_{\alpha}^1 + \varphi_{3\alpha}^0, \quad (20)$$

$$2\varepsilon_{\alpha 3}^1 = \varphi_{3\alpha}^1 - b_{\alpha}^{\delta} v_{\delta}^1. \quad (21)$$

Here, $\varphi_{\lambda\alpha}^n = v_{\lambda,\alpha}^n - b_{\lambda\alpha}^n v_3^n$, $\varphi_{3\alpha}^n = v_{3,\alpha}^n + b_{\alpha}^{\delta} v_{\delta}^n$, where $'$, α' denotes the partial derivatives of the coordinate axes, $'n'$ takes 0 or 1, representing translational and rotational parameters, respectively. Furthermore, $b_{\lambda\alpha}$ and b_{α}^{δ} denote the covariant and mixed-variant components of curvature tensors. For plate structures, $b_{\lambda\alpha} = b_{\alpha}^{\delta} = 0$. Thus, The strain-displacement relationships for the MEE plates can be expressed as

$$\varepsilon_{11} = v_{1,1}^0 + \Theta^3 v_{1,1}^1, \quad (22)$$

$$\varepsilon_{22} = v_{2,2}^0 + \Theta^3 v_{2,2}^1, \quad (23)$$

$$2\varepsilon_{12} = v_{1,2}^0 + v_{2,1}^0 + \Theta^3 (v_{1,2}^1 + v_{2,1}^1), \quad (24)$$

$$2\varepsilon_{23} = v_2^1 + v_{3,2}^0, \quad (25)$$

$$2\varepsilon_{13} = v_1^1 + v_{3,1}^0. \quad (26)$$

3 Finite element formulations

The MEE plate is discretized using an 8-node isoparametric element, with each node possessing 5 displacement DOFs ($u, v, w, \varphi_1, \varphi_2$). Each MEE layer in an element has one electric potential difference DOF ϕ , one magnetic potential difference DOF ψ and one temperature difference DOF θ . Since the elements in the curvilinear coordinate system (Θ^1, Θ^2) are irregularly shaped, a coordinate

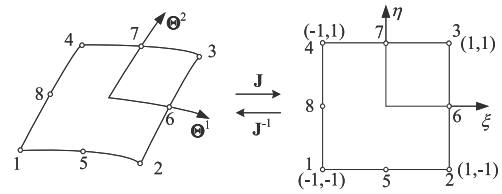


Fig. 2 Schematic of eight-node element

transformation as illustrated in Fig. 2 is created to transform them into regularly shaped elements in the natural coordinate system (ξ, η).

The displacement vector $\mathbf{v}$ and strain vector $\boldsymbol{\varepsilon}$ are formulated in terms of the nodal degrees of freedom vector $\mathbf{q}$ and the interpolation functions

$$\mathbf{v} = \mathbf{N}\mathbf{q}. \quad (27)$$

$$\boldsymbol{\varepsilon} = \mathbf{L}\mathbf{v} = \mathbf{L}\mathbf{N}\mathbf{q} = \mathbf{B}_u\mathbf{q}. \quad (28)$$

Here, $\mathbf{B}_u$ is the strain field matrix, $\mathbf{L}$ denotes the differential operator. $\mathbf{N}$ is the shape function, which is also the interpolation function defined by the natural coordinates, expressed at each node of an element as

$$\begin{cases} N_i = \frac{1}{4}(1 + \xi_i\xi)(1 + \eta_i\eta)(\xi_i\xi + \eta_i\eta - 1) & \text{for } i = 1, 2, 3, 4 \\ N_i = \frac{1}{4}(1 - \xi^2)(1 + \eta_i\eta) & \text{for } i = 5, 7 \\ N_i = \frac{1}{4}(1 - \eta^2)(1 + \xi_i\xi) & \text{for } i = 6, 8 \end{cases}. \quad (29)$$

The partial derivative of the interpolation functions N_i with respect to ξ, η can be represented in matrix form as

$$\left\{ \begin{matrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{matrix} \right\} = \begin{bmatrix} \frac{\partial \Theta^1}{\partial \xi} & \frac{\partial \Theta^2}{\partial \xi} \\ \frac{\partial \Theta^1}{\partial \eta} & \frac{\partial \Theta^2}{\partial \eta} \end{bmatrix} \left\{ \begin{matrix} \frac{\partial N_i}{\partial \Theta^1} \\ \frac{\partial N_i}{\partial \Theta^2} \end{matrix} \right\} = \mathbf{J} \left\{ \begin{matrix} \frac{\partial N_i}{\partial \Theta^1} \\ \frac{\partial N_i}{\partial \Theta^2} \end{matrix} \right\}. \quad (30)$$

Here, $\mathbf{J}$ is the Jacobian matrix. The transformation relationship between the curvilinear coordinate system (Θ^1, Θ^2) and the natural coordinate system (ξ, η) can be expressed as

$$\left\{ \begin{matrix} \frac{\partial N_i}{\partial \Theta^1} \\ \frac{\partial N_i}{\partial \Theta^2} \end{matrix} \right\} = \mathbf{J}^{-1} \left\{ \begin{matrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{matrix} \right\} = \frac{1}{|\mathbf{J}|} \begin{bmatrix} \frac{\partial \Theta^2}{\partial \eta} & -\frac{\partial \Theta^2}{\partial \xi} \\ -\frac{\partial \Theta^1}{\partial \eta} & \frac{\partial \Theta^1}{\partial \xi} \end{bmatrix} \left\{ \begin{matrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{matrix} \right\}. \quad (31)$$

Derive the finite element dynamics equations using the Hamilton's principle as

$$\int_{t_1}^{t_2} (\delta T - \delta W_{\text{int}} + \delta W_{\text{ext}}) dt = 0. \quad (32)$$

Here δ denotes the variational operator, T , W_{int} and W_{ext} represent the kinetic energy, internal work and external work, respectively.

The expression for δT is given by

$$\delta T = - \int_V \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} dV = - \delta \mathbf{q}^T \mathbf{M}_{\text{uu}} \ddot{\mathbf{q}}, \quad (33)$$

with

$$\mathbf{M}_{\text{uu}} = \int_V \rho \mathbf{N}^T \mathbf{Z}_{\text{u}}^T \mathbf{Z}_{\text{u}} \mathbf{N} dV. \quad (34)$$

Here, ρ is the material density, $\mathbf{M}_{\text{uu}}$ is the mass matrix, $\ddot{\mathbf{q}}$ is the second derivative of nodal DOFs with respect to time.

From Eqs. 1–4, the expression for δW_{int} can be written as

$$\begin{aligned} \delta W_{\text{int}} &= \int_V (\delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} - \delta \mathbf{E}^T \mathbf{D} - \delta \mathbf{H}^T \mathbf{B} - \delta \boldsymbol{\theta}^T \mathbf{S}) dV \\ &= \delta \mathbf{q}^T (\mathbf{K}_{\text{uu}} \mathbf{q} + \mathbf{K}_{\text{u}\phi} \boldsymbol{\phi} + \mathbf{K}_{\text{u}\psi} \boldsymbol{\psi} + \mathbf{K}_{\text{uT}} \boldsymbol{\theta}) + \delta \boldsymbol{\phi}^T (\mathbf{K}_{\phi\text{u}} \mathbf{q} + \mathbf{K}_{\phi\phi} \boldsymbol{\phi} + \mathbf{K}_{\phi\psi} \boldsymbol{\psi} + \mathbf{K}_{\phi\text{T}} \boldsymbol{\theta}) \\ &\quad + \delta \boldsymbol{\psi}^T (\mathbf{K}_{\psi\text{u}} \mathbf{q} + \mathbf{K}_{\psi\phi} \boldsymbol{\phi} + \mathbf{K}_{\psi\psi} \boldsymbol{\psi} + \mathbf{K}_{\psi\text{T}} \boldsymbol{\theta}) + \delta \boldsymbol{\theta}^T (\mathbf{K}_{\text{Tu}} \mathbf{q} + \mathbf{K}_{\text{T}\phi} \boldsymbol{\phi} + \mathbf{K}_{\text{T}\psi} \boldsymbol{\psi} + \mathbf{K}_{\text{TT}} \boldsymbol{\theta}), \end{aligned} \quad (35)$$

with

$$\mathbf{K}_{\text{uu}} = \int_V \mathbf{B}_{\text{u}}^T \mathbf{C} \mathbf{B}_{\text{u}} dV, \quad (36)$$

$$\mathbf{K}_{\text{u}\phi} = \mathbf{K}_{\phi\text{u}}^T = - \int_V \mathbf{B}_{\text{u}}^T \mathbf{e}^T \mathbf{B}_{\phi} dV, \quad (37)$$

$$\mathbf{K}_{\text{u}\psi} = \mathbf{K}_{\psi\text{u}}^T = - \int_V \mathbf{B}_{\text{u}}^T \mathbf{q}^T \mathbf{B}_{\psi} dV, \quad (38)$$

$$\mathbf{K}_{\text{uT}} = \mathbf{K}_{\text{Tu}}^T = - \int_V \mathbf{B}_{\text{u}}^T \boldsymbol{\lambda}^T \mathbf{B}_{\text{T}} dV, \quad (39)$$

$$\mathbf{K}_{\phi\phi} = - \int_V \mathbf{B}_{\phi}^T \boldsymbol{\chi} \mathbf{B}_{\phi} dV, \quad (40)$$

$$\mathbf{K}_{\phi\psi} = \mathbf{K}_{\psi\phi}^T = - \int_V \mathbf{B}_{\phi}^T \boldsymbol{\kappa} \mathbf{B}_{\psi} dV, \quad (41)$$

$$\mathbf{K}_{\phi\text{T}} = \mathbf{K}_{\text{T}\phi}^T = - \int_V \mathbf{B}_{\phi}^T \boldsymbol{p}^T \mathbf{B}_{\text{T}} dV, \quad (42)$$

$$\mathbf{K}_{\psi\psi} = - \int_V \mathbf{B}_{\psi}^T \boldsymbol{\gamma} \mathbf{B}_{\psi} dV, \quad (43)$$

$$\mathbf{K}_{\psi\text{T}} = \mathbf{K}_{\text{T}\psi}^T = - \int_V \mathbf{B}_{\psi}^T \boldsymbol{\tau}^T \mathbf{B}_{\text{T}} dV, \quad (44)$$

$$\mathbf{K}_{\text{TT}} = - \int_V \mathbf{B}_{\text{T}}^T \mathbf{c} \mathbf{B}_{\text{T}} dV. \quad (45)$$

Here, $\mathbf{K}_{\text{uu}}$, $\mathbf{K}_{\text{u}\phi}$, $\mathbf{K}_{\text{u}\psi}$, $\mathbf{K}_{\text{uT}}$, $\mathbf{K}_{\phi\phi}$, $\mathbf{K}_{\phi\psi}$, $\mathbf{K}_{\phi\text{T}}$, $\mathbf{K}_{\psi\psi}$, $\mathbf{K}_{\psi\text{T}}$ and $\mathbf{K}_{\text{TT}}$ represent the elastic stiffness matrix, coupled electro-elastic stiffness matrix, coupled magneto-elastic stiffness matrix, coupled thermo-elastic stiffness matrix, electric stiffness matrix, coupled electro-magnetic stiffness matrix, coupled thermo-electric stiffness matrix, magnetic stiffness matrix, coupled thermo-magnetic stiffness matrix, and thermal stiffness matrix, respectively.

The external work can be written as

$$\begin{aligned} \delta W_{\text{ext}} &= \int_V \delta \mathbf{u}^T \mathbf{f}_{\text{b}} dV + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f}_{\text{s}} d\Omega + \delta \mathbf{u}^T \mathbf{f}_{\text{c}} \\ &\quad - \int_{\Omega} \delta \boldsymbol{\phi}^T \boldsymbol{\vartheta} d\Omega - \delta \boldsymbol{\phi}^T \mathbf{Q}_{\text{c}} - \int_{\Omega} \delta \boldsymbol{\psi}^T \mathbf{i} d\Omega - \delta \boldsymbol{\psi}^T \mathbf{I}_{\text{c}} - \int_{\Omega} \delta \boldsymbol{\theta}^T \bar{\mathbf{Q}} d\Omega \\ &= \delta \mathbf{q}^T \mathbf{F}_{\text{ue}} + \delta \boldsymbol{\phi}^T \mathbf{F}_{\phi\text{e}} + \delta \boldsymbol{\psi}^T \mathbf{F}_{\psi\text{e}} + \delta \boldsymbol{\theta}^T \mathbf{F}_{\text{Te}}, \end{aligned} \quad (46)$$

with

$$\mathbf{F}_{ue} = \int_V \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_b dV + \int_{\Omega} \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_s d\Omega + \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_c, \quad (47)$$

$$\mathbf{F}_{\phi e} = - \int_{\Omega} \boldsymbol{\vartheta} d\Omega - \mathbf{Q}_c, \quad (48)$$

$$\mathbf{F}_{\psi e} = - \int_{\Omega} \boldsymbol{\iota} d\Omega - \mathbf{I}_c, \quad (49)$$

$$\mathbf{F}_{Te} = - \int_{\Omega} \bar{\mathbf{Q}} d\Omega. \quad (50)$$

Here, $\mathbf{f}_b$, $\mathbf{f}_s$ and $\mathbf{f}_c$ are the body force, surface traction and concentrated point force vectors, respectively. Moreover, $\boldsymbol{\vartheta}$ and $\mathbf{Q}_c$ are the vectors representing the homogeneous and concentrated charges, $\boldsymbol{\iota}$ and $\mathbf{I}_c$ are the homogeneous and concentrated current vectors, and $\bar{\mathbf{Q}}$ is the homogeneous heat flow vector. Furthermore, $\mathbf{F}_{ue}$, $\mathbf{F}_{\phi e}$, $\mathbf{F}_{\psi e}$ and $\mathbf{F}_{Te}$ represent the applied mechanical load, electric load, magnetic load, and thermal load vectors, respectively.

Substituting Eqs. 33, 35 and 46 into 32 yields the dynamic FE equations as

$$\mathbf{M}_{uu} \ddot{\mathbf{q}} + \mathbf{K}_{uu} \mathbf{q} + \mathbf{K}_{u\phi} \boldsymbol{\phi}_a + \mathbf{K}_{u\psi} \boldsymbol{\psi}_a + \mathbf{K}_{uT} \boldsymbol{\theta}_a = \mathbf{F}_{ue}, \quad (51)$$

$$\mathbf{K}_{\phi u} \mathbf{q} + \mathbf{K}_{\phi\phi} \boldsymbol{\phi}_s + \mathbf{K}_{\phi\psi} \boldsymbol{\psi}_s + \mathbf{K}_{\phi T} \boldsymbol{\theta}_s = \mathbf{F}_{\phi e}, \quad (52)$$

$$\mathbf{K}_{\psi u} \mathbf{q} + \mathbf{K}_{\psi\phi} \boldsymbol{\phi}_s + \mathbf{K}_{\psi\psi} \boldsymbol{\psi}_s + \mathbf{K}_{\psi T} \boldsymbol{\theta}_s = \mathbf{F}_{\psi e}, \quad (53)$$

$$\mathbf{K}_{Tu} \mathbf{q} + \mathbf{K}_{T\phi} \boldsymbol{\phi}_s + \mathbf{K}_{T\psi} \boldsymbol{\psi}_s + \mathbf{K}_{TT} \boldsymbol{\theta}_s = \mathbf{F}_{Te}. \quad (54)$$

In the equations, $\boldsymbol{\theta}_s$ physically represents the temperature field variation induced by structural deformation under external loads, which is the primary focus of this study.

By removing the dynamic part, i.e., eliminating the terms containing $\ddot{\mathbf{q}}$, the static FE equations are obtained.

4 Numerical analysis

This section presents the numerical validation of the fully coupled FE model, comparing the influence of the multiphysical fields on the temperature field with

COMSOL results, and analyzing the impact of the boundary conditions on the output.

In COMSOL, the solid mechanics and heat transfer modules treat thermal loads merely as inputs to compute thermal expansion, which does not allow the establishment of the fully coupled model proposed in this study or the realization of temperature field variations within the structure. Therefore, the built-in piezoelectric module was employed instead. In this module, applying mechanical loads yields both the displacement field and the electric field distribution of the structure. By equivalently treating thermal stress parameters as piezoelectric coefficients and heat capacity as dielectric constants within the material properties of COMSOL, the resulting electric field can then be interpreted as the structural temperature field variation.

During the model validation with COMSOL, the applied voltage and magnetic potential require conversion into equivalent stress, whereas the present model allows direct application without such conversion.

4.1 Mesh convergence analysis

A cantilevered square plate is considered, whose geometric configuration is depicted in Fig. 3 and material parameters are summarized in Table 1. The material properties of the MEE composite are obtained using the rule of mixtures homogenization scheme, which combines the properties of the ferroelectric phase (BaTiO_3) and ferromagnetic (CoFe_2O_4) phase according to their respective volume fractions. Specifically,

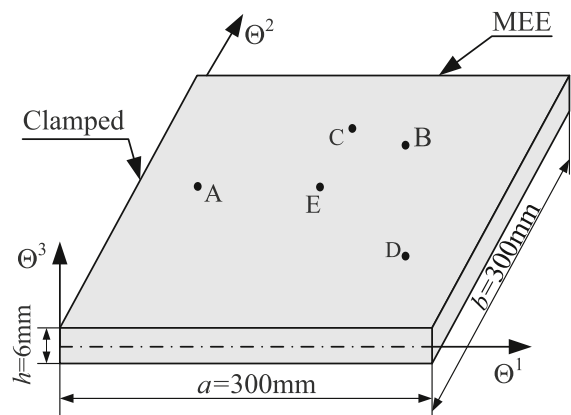


Fig. 3 MEE square plate

Table 1 Material parameters of the MEE plate (Kondaiah et al. 2012; Kim 2011)

Properties	MEE	CoFe ₂ O ₄	BaTiO ₃	0.5V _f (Sladek et al. 2013)
$C_{11} = C_{22}$ (GPa)	200	286	166	213
C_{12} (GPa)	110	173	77	113
$C_{13} = C_{23}$ (GPa)	110	170	78	0
C_{33} (GPa)	190	269.5	162	207
$C_{44} = C_{55}$ (GPa)	45	45.3	43	49.9
C_{66} (GPa)	45	56.5	44.5	50
$e_{31} = e_{32}$ (C/m ²)	− 3.5	0	− 4.4	− 2.71
e_{33} (C/m ²)	11.0	0	18.6	8.86
$e_{24} = e_{15}$ (C/m ²)	0	0	11.6	0.15
$\chi_{11} = \chi_{22}$ (10 ^{−9} C ² /Nm ²)	0.9	0.08	11.2	0.24
χ_{33} (10 ^{−9} C ² /Nm ²)	7.5	0.093	12.6	6.37
$q_{31} = q_{32}$ (N/Am)	200	580	0	222
q_{33} (N/Am)	260	700	0	292
$q_{24} = q_{15}$ (N/Am)	180	560	0	185
$\kappa_{11} = \kappa_{22}$ (10 ^{−12} Ns/VC)	6.0	0	0	− 5.23
κ_{33} (10 ^{−12} Ns/VC)	2500	0	0	2750
$\gamma_{11} = \gamma_{22}$ (10 ^{−4} Ns ² /C ²)	− 1.5	− 5.9	0.05	2.01
γ_{33} (10 ^{−4} Ns ² /C ²)	0.75	1.57	0.1	0.839
$\alpha_1 = \alpha_2$ (10 ^{−6} 1/K)	12.9	10	15.7	−
α_3 (10 ^{−6} 1/K)	7.8	10	6.4	−
p (10 ^{−7} C/m ² K)	− 9	0	0	−
τ (10 ^{−5} C/m ² K)	− 18	0	0	−
c (J/m ³ K)	327	165	422	−
ρ (kg/m ³)	5600	5300	5800	5550

for a generic material property P , the effective value is calculated as:

$$P = v_f P_f + v_m P_m, \quad (55)$$

where v_f and v_m are the volume fractions of the ferroelectric and ferromagnetic phases, respectively, and

P_f and P_m are the corresponding material parameters of each phase.

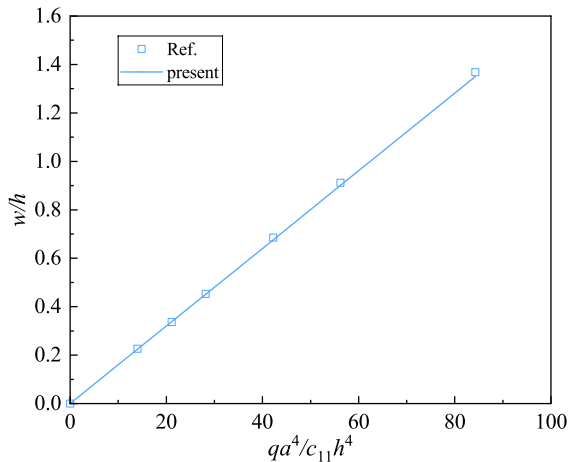
The present FE model utilizes shell elements, while COMSOL employs 3D solid elements. The frequencies and the midpoint displacement at the free end under a uniform surface load of 15,000 Pa

Table 2 The first-order frequency and displacements vs. mesh density (COMSOL model)

Mesh	5×5×10	10×10×10	20×20×10	30×30×10	40×40×10	45×45×10	50×50×10
DOFs	6789	24144	90804	200064	351924	443829	546384
Frequency(Hz)	56.398	55.152	54.637	54.493	54.428	54.407	54.391
Diff.(%)		− 2.21	− 0.93	− 0.26	− 0.12	− 0.04	− 0.03
Displacements(mm)	− 5.5634	− 5.7984	− 5.9036	− 5.9342	− 5.9484	− 5.9525	− 5.9564
Diff.(%)		4.22	1.81	0.52	0.24	0.08	0.06

Table 3 The first-order frequency and displacements vs. mesh density (present model)

Mesh	5×5	6×6	8×8	10×10	12×12	14×14	16×16	18×18	20×20
DOFs	675	960	1680	2600	3720	5040	6560	8280	10200
Frequency(Hz)	52.252	52.228	52.209	52.202	52.199	52.198	52.197	52.197	52.196
Diff.(%)		− 0.05	− 0.04	− 0.01	− 0.01	0.00	0.00	0.00	0.00
Displacements(mm)	− 5.9941	− 5.9976	− 5.9993	− 5.9994	− 5.9992	− 5.9989	− 5.9987	− 5.9985	− 5.9984
Diff.(%)		0.06	0.03	0.00	0.00	− 0.01	0.00	0.00	0.00

**Fig. 4** Variation of the central deflection with the load intensity for the clamped plate

of COMSOL model and the present model, using different mesh methods, are separately investigated and the results are presented in Tables 2 and 3. The outcomes show that the first-order frequencies and

the midpoint displacements of COMSOL model and the present model gradually converge to a constant value as the number of meshes increases. The present model achieves convergence at a first-order frequency of 52.228 Hz and a displacement of −5.9976 mm with the required mesh of 6×6 elements (960 DOFs), while the COMSOL model converges at 54.407 Hz and −5.9525 mm with the required mesh of 45×45×10 elements (443829 DOFs). That is, the present model only needs 0.22% DOFs of COMSOL for convergence, with the deviations in the first-order frequency and displacement are merely 4.00% and 0.76%, respectively.

4.2 Model validation

The simulation case considers a square MEE plate with all four edges clamped, based on the model reported by Sladek et al. (2013). The plate has a length and width of $a = b = 254$ mm, a thickness of $h = 12$ mm, and the material properties corresponding to 0.5 volume fraction of BaTiO₃/CoFe₂O₄ composite, as listed in Table 1. A normalized uniform

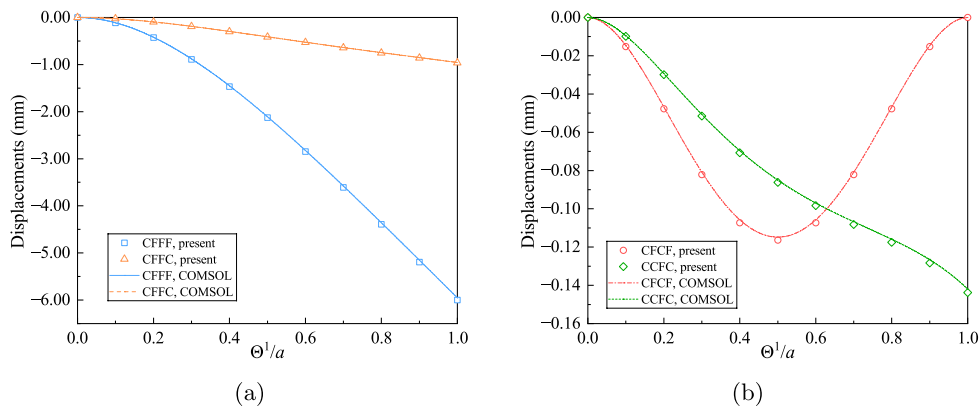
**Fig. 5** Vertical displacements of the center-line under mechanical loads **a** CFFF and CFEC, **b** CFCE and CCFC

Fig. 6 map under CFFF condition and mechanical loads **a** present, **b** COM-SOL

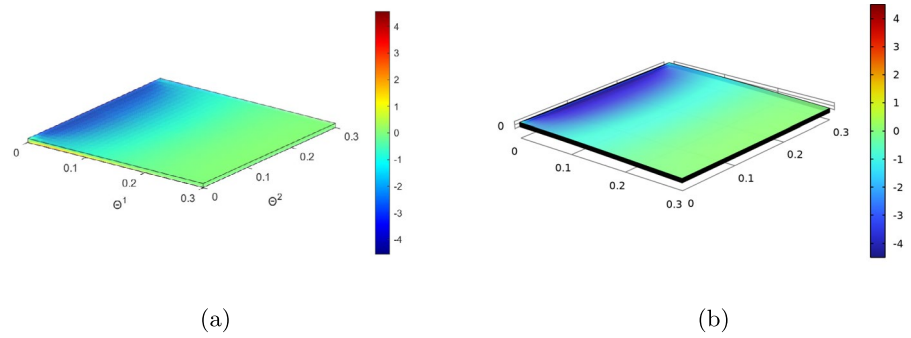


Fig. 7 Temperature contour map under CFCF condition and mechanical loads **a** present, **b** COMSOL

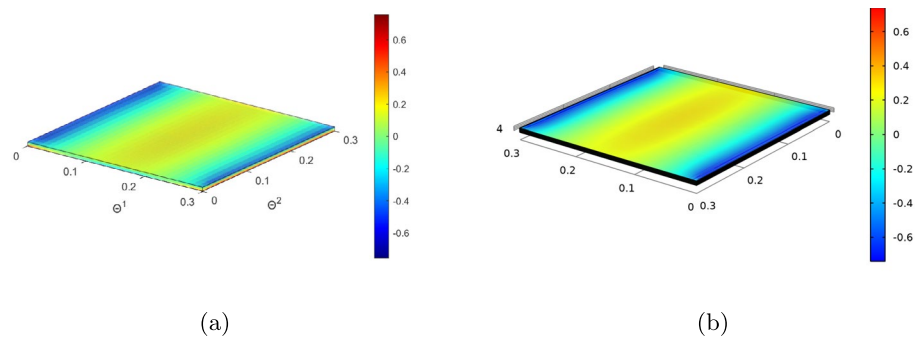


Fig. 8 Temperature contour map under CFCC condition and mechanical loads **a** present, **b** COMSOL

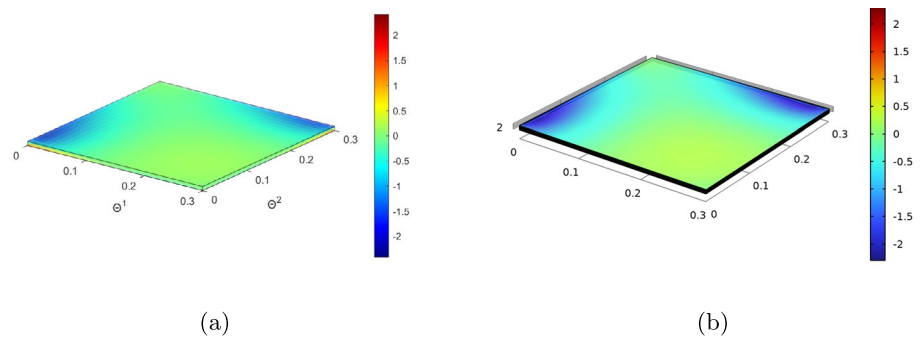
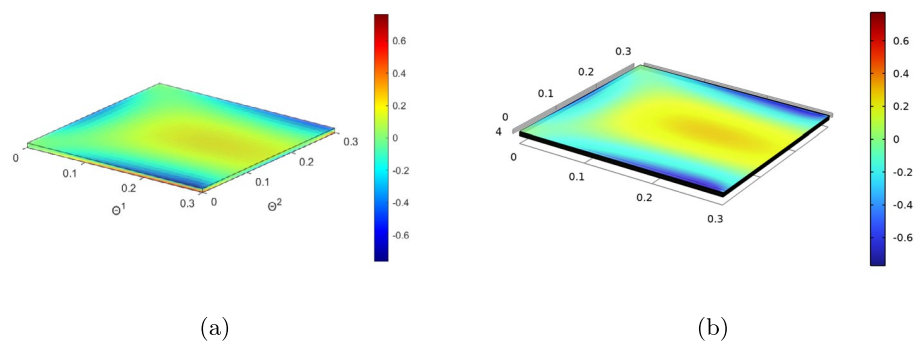


Fig. 9 Temperature contour map under CCFC condition and mechanical loads **a** present, **b** COMSOL



load $\tilde{q} = qa^4/C_{11}h^4$ is applied on the upper surface of the plate. Figure 4 shows the variation of the dimensionless central deflection with respect to the load intensity. The comparison indicates that the present model exhibits good agreement with the reference results, thereby establishing a solid theoretical basis for subsequent investigations of temperature field variations induced by structural deformation.

4.3 Static studies of MEE plate

The static response of a square MEE plate is investigated. Four points are considered for evaluation, namely A(0.055m, 0.155m), B(0.205m, 0.205m), C(0.155m, 0.235m), D(0.255m, 0.085m). The dimensions of the plate and the positions of these points are shown in Fig. 3. The corresponding material parameters are summarized in Table 1. Four types of boundary conditions are considered: CFFF, CFCF, CFFC, and CCFC, in which C and F denote "Clamp" and "Free". The sequence of C and F corresponds to the boundary conditions on the left, bottom, right, and top edges of the plate, respectively. The present model and COMSOL are used to calculate the output temperature difference θ along the thickness corresponding to these points under the impact of different physical fields, respectively. The meshing of the present model is 30×30, while COMSOL uses solid elements with 50×50×10 elements.

4.3.1 Elasto-thermal bidirectional coupling

Applying a uniform pressure of 1.5×10^4 Pa on the top surface of the plate, the vertical displacements of the center-line along the Θ^1 -direction of the plate are calculated and shown in Fig. 5. The temperature difference distribution cloud diagrams under different boundary conditions are illustrated in Figs. 6, 7, 8 and 9. There is a high degree of consistency between the present model's outcomes and COMSOL simulations. Taking the CFFF condition as an example, the present model only has 22800 DOFs, accounting for approximately 4.17% of COMSOL model's DOF, which illustrates high accuracy and efficiency of the current model. Furthermore, the boundary conditions exert a substantial effect on the plate response: weaker constraints lead to larger deformations, with the CFFF case exhibiting the greatest displacement amplitude (Fig. 5). Larger deformations also amplify the thermo-mechanical coupling at the clamped edge. As illustrated in Figs. 6, 7, 8 and 9, under the CFFF condition, the temperature difference at the clamped edge reaches its maximum. This arises because the constrained region must resist the deformation, thereby developing the highest stresses and most pronounced coupling effects.

4.3.2 Electro-thermal bidirectional coupling

Simultaneously applying voltages of 300V and -300V on the top and the bottom surfaces of the plate, the vertical displacements of the center-line in the Θ^1

Table 4 Vertical displacements of the center-line in the Θ^1 -direction under electric loads (mm)

Θ^1/a	CFFF		CFCF		CFFC		CCFC	
	present	COMSOL	present	COMSOL	present	COMSOL	present	COMSOL
0.00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.10	-1.65E-03	-1.61E-03	7.29E-04	7.05E-04	3.29E-05	3.46E-05	1.87E-04	1.81E-04
0.20	-7.27E-03	-7.19E-03	2.13E-03	2.10E-03	-4.80E-04	-4.62E-04	6.24E-04	6.12E-04
0.30	-1.75E-02	-1.74E-02	3.45E-03	3.41E-03	-2.07E-03	-2.03E-03	1.17E-03	1.15E-03
0.40	-3.25E-02	-3.24E-02	4.31E-03	4.28E-03	-5.01E-03	-4.94E-03	1.68E-03	1.67E-03
0.50	-5.23E-02	-5.23E-02	4.61E-03	4.58E-03	-9.46E-03	-9.34E-03	1.95E-03	1.93E-03
0.60	-7.67E-02	-7.70E-02	4.31E-03	4.28E-03	-1.56E-02	-1.55E-02	1.60E-03	1.60E-03
0.70	-1.06E-01	-1.05E-01	3.45E-03	3.41E-03	-2.39E-02	-2.37E-02	7.30E-05	9.41E-05
0.80	-1.39E-01	-1.39E-01	2.13E-03	2.10E-03	-3.48E-02	-3.45E-02	-3.54E-03	-3.48E-03
0.90	-1.77E-01	-1.77E-01	7.29E-04	7.05E-04	-4.90E-02	-4.87E-02	-1.04E-02	-1.03E-02
1.00	-2.19E-01	-2.18E-01	0.00E+00	0.00E+00	-6.74E-02	-6.70E-02	-2.21E-02	-2.20E-02

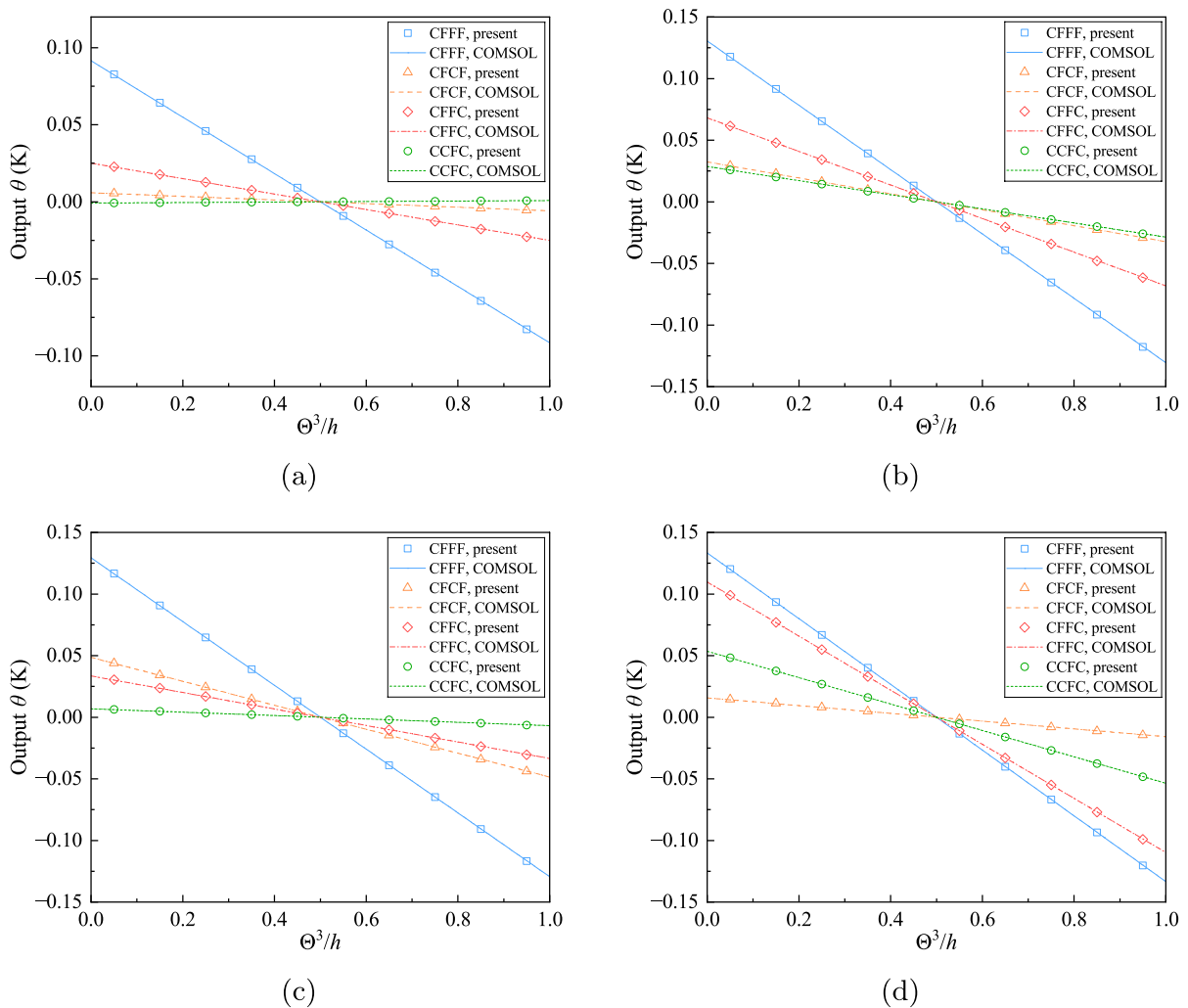


Fig. 10 Output temperature difference θ in thickness direction under electric loads **a** at point A, **b** at point B, **c** at point C, **d** at point D

-direction of the plate are calculated, with the results shown in Table 4. The output temperature differences in the Θ^3 -direction at points A-D under various boundary conditions are illustrated in Fig. 10. The results confirm that the center-line displacements and temperature differences determined by the present model exhibit strong consistency with the COMSOL simulations. From Table 4, it can be seen that under the electric field, the effects of CFFF and CFCC conditions on the center-line displacements remain unchanged, while under CFCF condition, the center-line displacement is positive, and under CCFC condition, the center-line displacement changes from positive to negative along the Θ^1 -direction.

The displacement under electric loads differs from that under mechanical loads because, in terms of equivalent stress, the electric loads induces equivalent stresses in the Θ^1 and Θ^2 -directions, whereas the mechanical loads induces equivalent stress only in the Θ^3 -direction. As evidenced in Fig. 10, the temperature differences in the Θ^3 -direction at different points are related to both the boundary conditions and the point locations. This can be attributed to the variations in deformation at different positions of the plate under different boundary conditions, which in turn lead to differences in temperature.

Table 5 Vertical displacements of the center-line in the Θ^1 -direction under magnetic loads (mm)

Θ^1/a	CFFF		CFCF		CFFC		CCFC	
	present	COMSOL	present	COMSOL	present	COMSOL	present	COMSOL
0.00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.10	5.50E-03	5.32E-03	-2.43E-03	-2.37E-03	-2.91E-04	-1.16E-04	-6.26E-04	-6.05E-04
0.20	2.43E-02	2.39E-02	-7.12E-03	-7.01E-03	1.11E-03	1.54E-03	-2.08E-03	-2.05E-03
0.30	5.84E-02	5.78E-02	-1.15E-02	-1.14E-02	6.26E-03	6.79E-03	-3.91E-03	-3.86E-03
0.40	1.09E-01	1.08E-01	-1.44E-02	-1.43E-02	1.62E-02	1.65E-02	-5.62E-03	-5.57E-03
0.50	1.75E-01	1.73E-01	-1.54E-02	-1.53E-02	3.15E-02	3.12E-02	-6.50E-03	-6.46E-03
0.60	2.56E-01	2.55E-01	-1.44E-02	-1.43E-02	5.32E-02	5.17E-02	-5.36E-03	-5.36E-03
0.70	3.53E-01	3.51E-01	-1.15E-02	-1.14E-02	8.26E-02	7.92E-02	-2.44E-04	-3.15E-04
0.80	4.65E-01	4.63E-01	-7.12E-03	-7.01E-03	1.22E-01	1.15E-01	1.18E-02	1.16E-02
0.90	5.91E-01	5.89E-01	-2.43E-03	-2.36E-03	1.73E-01	1.63E-01	3.49E-02	3.46E-02
1.00	7.32E-01	7.29E-01	0.00E+00	0.00E+00	2.40E-01	2.24E-01	7.40E-02	7.35E-02

4.3.3 Magneto-thermal bidirectional coupling

Magnetic potential loads of 200A and -200A are applied to the top and the bottom surfaces of the plate, respectively. The vertical displacements of the center-line along the Θ^1 -direction are calculated and shown in Table 5. The output temperature differences in the Θ^3 -direction at points A-D under different boundary conditions are illustrated in Fig. 11. According to Tables 4 and 5 and Figs. 10 and 11, the data reveal that compared to the electric field effect, the center-line displacements under magnetic field show opposite trends and larger amplitudes, which leads to the temperature differences exhibiting the same variation trend. This is because the piezomagnetic coefficients q_{31} and q_{32} are negative and have larger absolute values compared to the piezoelectric coefficients.

4.4 Dynamic analysis of MEE plate

The dynamic analysis of a square MEE plate is studied, the dimensions and materials are consistent with those in Section 4.3. Two different boundary conditions are considered: CFFF and CFCF. The meshing methods of the present model and COMSOL are the same as in Section 4.3. The plate is exposed to independent mechanical, electric, and magnetic field forces, the vibrations of the center point under various boundary conditions are investigated. Furthermore, to analyze the variation of dynamic temperature difference θ along the thickness direction, the plate is discretized into 10 layers

with equal thickness for calculation. The layers are numbered sequentially from 1 to 10, with the numbering beginning at the bottom and ending at the top. The point E (0.155m, 0.155m) is arbitrarily selected in the Θ^1 - Θ^2 plane, as shown in Fig. 3, and the temperature differences at each layer along the thickness direction at this point can be expressed as $\theta = \{\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7, \theta_8, \theta_9, \theta_{10}\}^T$.

4.4.1 Elasto-thermal bidirectional coupling

Applying a uniform load of 1.5×10^4 Pa on the upper surface of the plate. Under CFFF condition, the total time is set to 4×10^{-2} s, with time increments of 1×10^{-4} s; under CFCF condition, the total time is set to 6×10^{-3} s, with time increments of 1.5×10^{-5} s. The vibrations of the center point and the dynamic output temperature differences in the thickness direction at point E under two different boundary conditions are presented in Figs. 12 and 13.

The excellent agreement between the present model and COMSOL simulations, as evidenced in Figs. 12 and 13, validates the model's accuracy in predicting the dynamic response of MEE plates. Additionally, boundary conditions have a considerable impact on the transient response of MEE plates. Compared with CFFF condition, the dynamic displacements at the center point show smaller amplitudes under CFCF condition. The magnitude of the

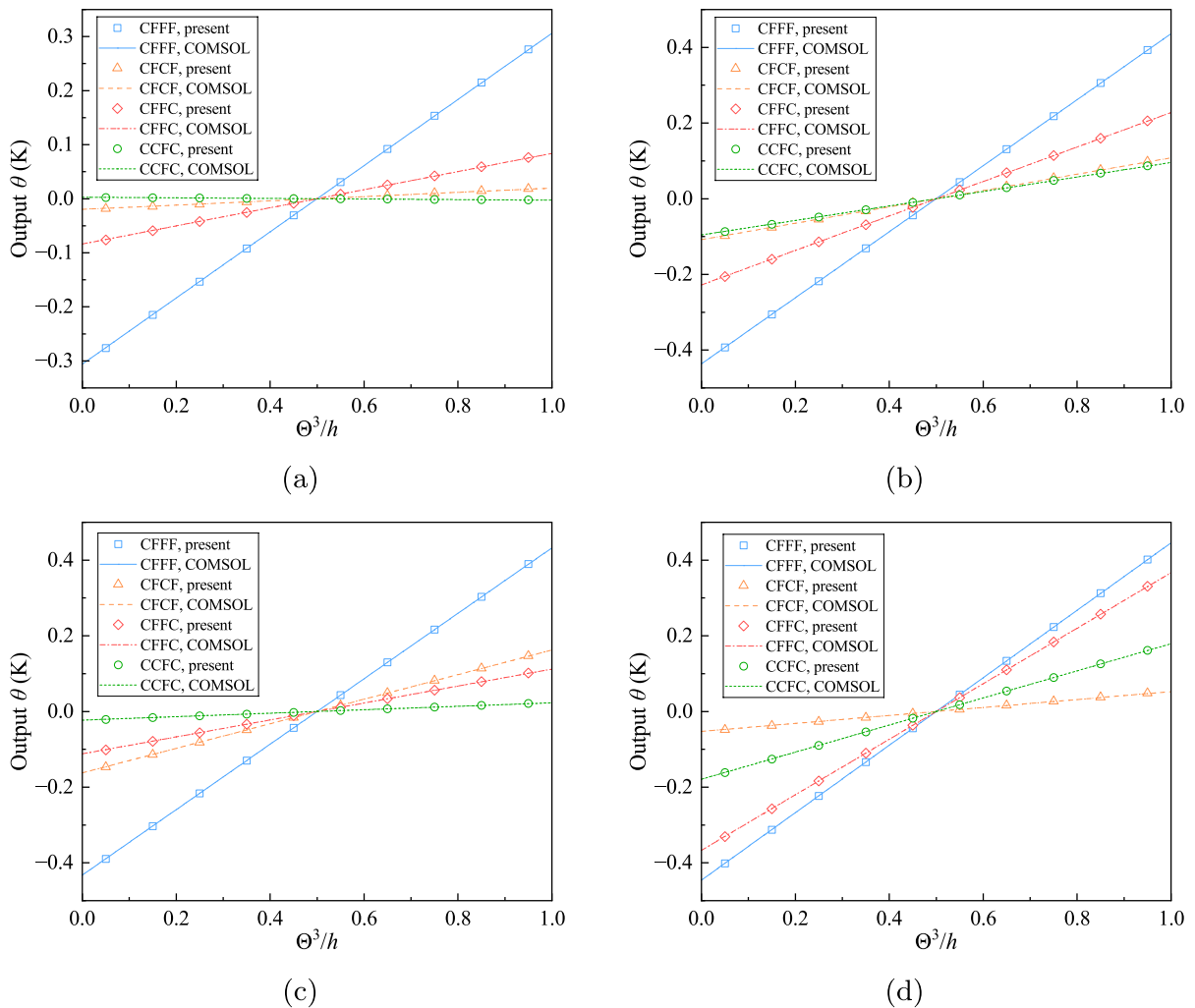


Fig. 11 Output temperature difference θ in thickness direction under magnetic loads **a** at point A, **b** at point B, **c** at point C, **d** at point D

dynamic temperature differences at point E is similarly reduced.

4.4.2 Electro-thermal bidirectional coupling

Simultaneously applying voltages of 300V and -300V on the plate's top and bottom surfaces. To more clearly present the comparison of temperature difference results between the present model and COMSOL, under CFFF condition, the total time is set to 7×10^{-4} s, with time increments of 1.4×10^{-5} s; under CCFC condition, the total time is set to 6.45×10^{-4} s, with time increments of 1.5×10^{-5} s. The vibrations of the center point and the dynamic

output temperature differences $\theta_1 - \theta_{10}$ at point E under two different boundary conditions are shown in Figs. 14 and 15. The figures show that under the electric field, the displacement variations at the center point and the temperature differences at point E exhibit more complex trends. Similarly, under CCFC condition, the dynamic displacements at the center point show smaller amplitudes compared to them under CFFF condition. The magnitude of dynamic temperature differences at point E are also reduced accordingly.

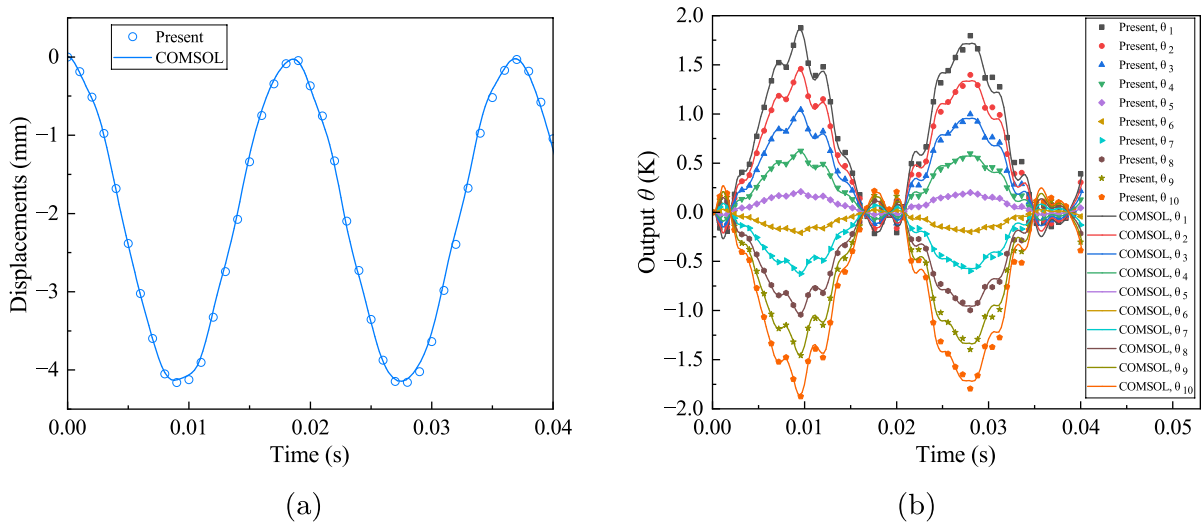


Fig. 12 Dynamic response under CFFF boundary condition and mechanical loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

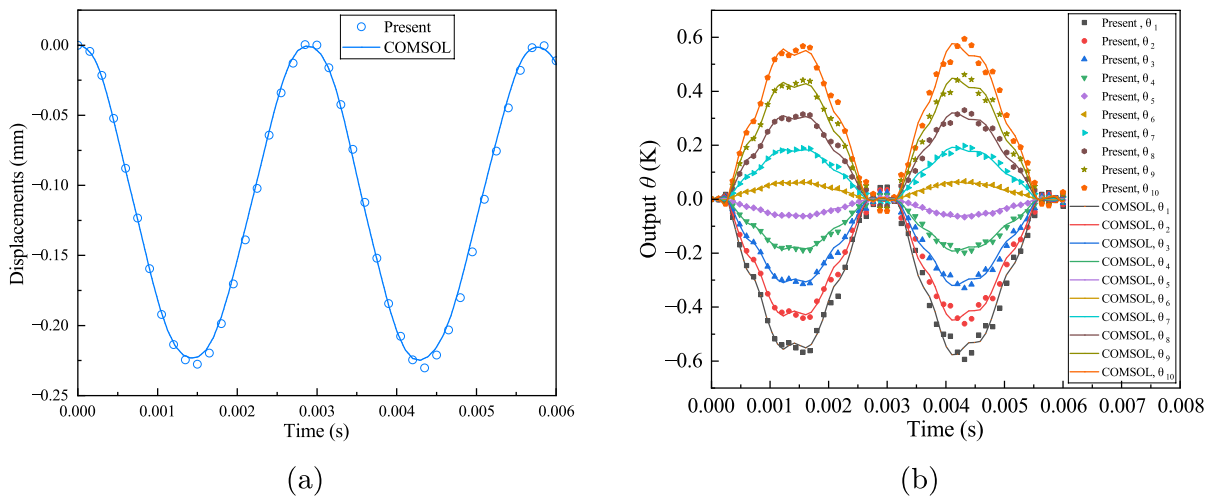


Fig. 13 Dynamic response under CFCF boundary condition and mechanical loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

4.4.3 Magneto-thermal bidirectional coupling

Magnetic potential loads of 200A and -200A are applied to the plate's upper and bottom surfaces, respectively. When comparing dynamic temperature differences, the total time is set to 6.6×10^{-4} s, with time increments of 3×10^{-6} s. The vibrations of the center point and the dynamic output temperature

differences in the thickness direction at point E under two different boundary conditions are shown in Figs. 16 and 17. By comparing the results presented in Figs. 14 and 15, it is obvious that the dynamic displacements and temperature differences at the same location show opposite trends and larger amplitudes, which is consistent with the static response and is also due to the effect of piezomagnetic coefficients.

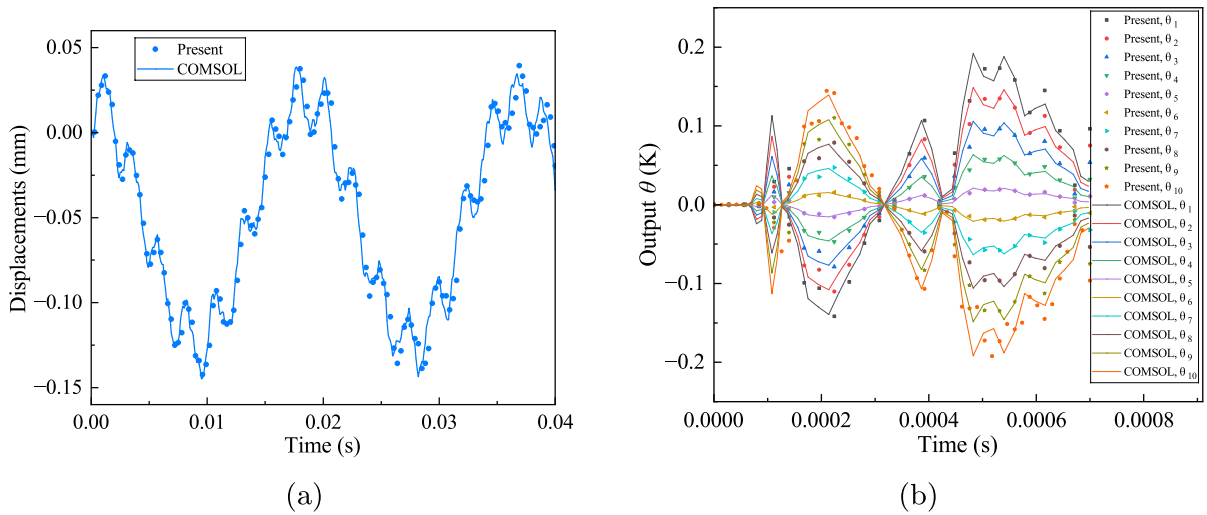


Fig. 14 Dynamic response under CFFF boundary condition and electric loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

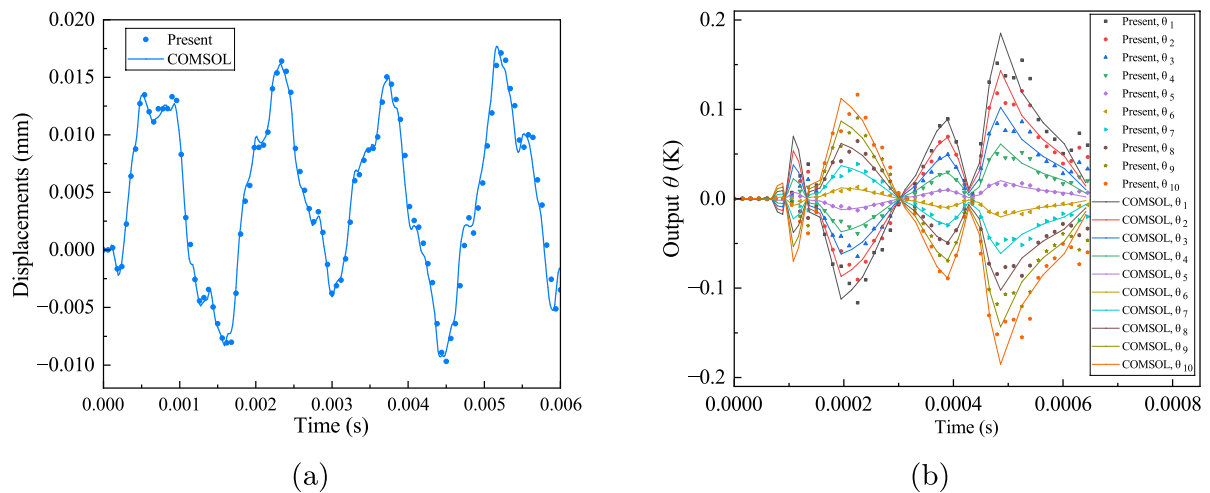


Fig. 15 Dynamic response under CFCF boundary condition and electric loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

5 Conclusions

Unlike most previous studies that treat temperature changes as a thermal load causing structural deformation, the innovation of this paper lies in focusing

on the temperature changes induced by structural deformation.

Based on the FOSD hypothesis, the present work develops a fully coupled finite element model of MEE plates and codes a corresponding computational

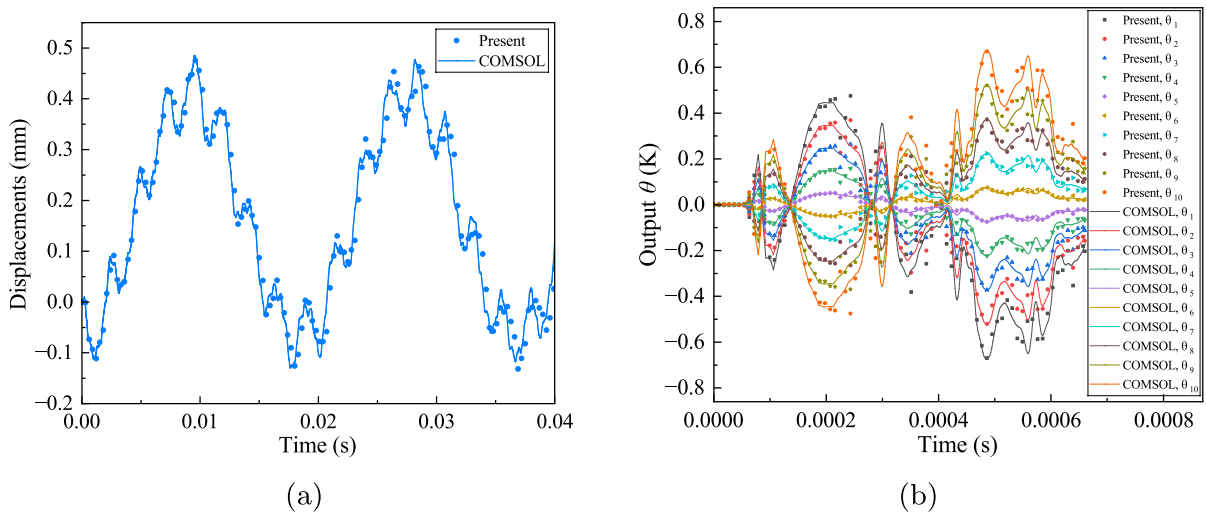


Fig. 16 Dynamic response under CFFF boundary condition and magnetic loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

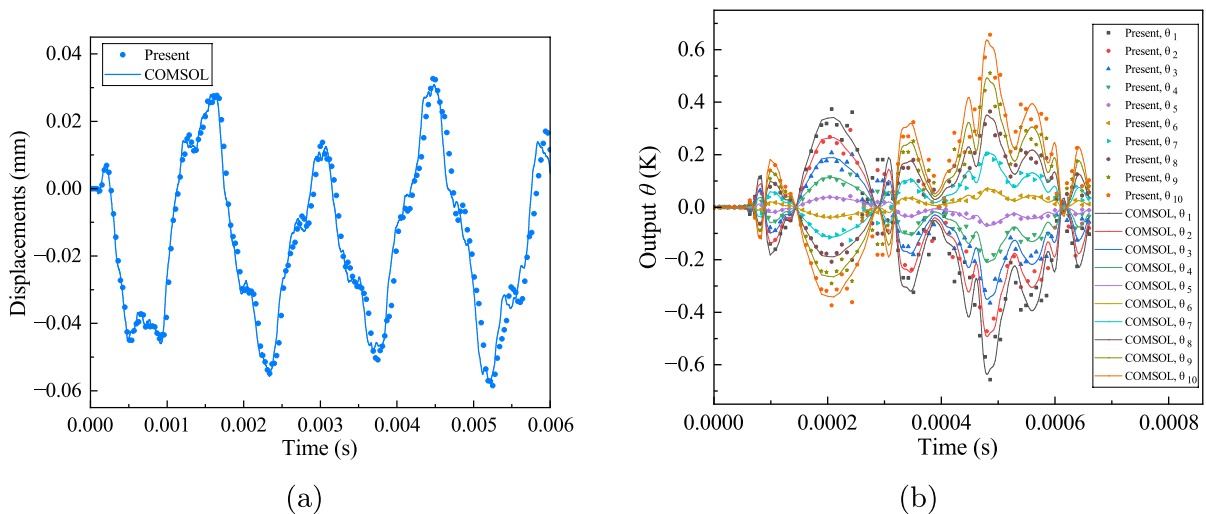


Fig. 17 Dynamic response under CFCF boundary condition and magnetic loads **a** dynamic displacements of the center point, **b** dynamic output temperature difference θ in the thickness direction at point E

program. The model analyzes the static bending and free vibration under multi-physics fields, explicitly accounting for displacement-induced temperature difference. After mesh convergence and validation, results are compared with COMSOL, showing strong agreement and confirming the model's accuracy. Moreover, the present model achieves comparable

precision with significantly fewer DOFs, manifesting its superior computational efficiency.

From the present results, it is evident that structural deformation does indeed lead to variations in the temperature field, an aspect overlooked in previous studies. The boundary conditions have a significant impact on the displacement and through-thickness temperature difference: the fewer fixed boundaries, the larger the deformation

of the plate, and the greater the amplitudes of the temperature difference. Secondly, the dynamic response and temperature field variations under electric and magnetic fields are more complex than under mechanical field. This is because the equivalent stresses induced by electric and magnetic fields act simultaneously in the Θ^1 and Θ^2 -directions, whereas the mechanical load acts only in the Θ^3 -direction. Moreover, the displacement and temperature difference variations under magnetic field show opposite trends and larger amplitudes compared to those under electric field, which is due to the opposite sign and larger magnitude of piezomagnetic coefficients compared to piezoelectric coefficients. Finally, most previous studies have focused only on the influence of physical fields on the displacement field of MEE plates. The present model further investigates the impact of displacement field on the temperature field, which has theoretical significance for the practical applications of MEE plates in real environments.

In this work, all parameters were treated as deterministic to focus on validating the fully coupled finite element model. For practical applications, however, uncertainties in material properties and external loading may significantly affect the response. Future work will consider incorporating variance-based uncertainty quantification and explore advanced machine learning frameworks such as the Deep Energy Method, systematically assess the influence of uncertain inputs on displacement and field distributions. While such advanced frameworks are beyond the scope of the current work, the present FEM-based formulation provides a solid foundation upon which uncertainty analyses can be systematically developed.

Author contributions S.Z. designed the study, developed the methodology, and provided resources. S.Q. implemented the software, performed validation, and wrote the initial draft. S.L. conducted investigations and visualized the results. C.L. supervised the research. S.Z. and S.M. contributed to manuscript review and editing.

Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Appendix A: Lengthy constitutive equations and detailed mathematical formulations

Constitutive equations for magneto-electro-elastic materials are as follows

$$\begin{aligned} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{Bmatrix} &= \begin{bmatrix} C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{22} & 0 & 0 & 0 \\ 0 & 0 & C_{66} & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & C_{55} \end{bmatrix} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{Bmatrix} - \begin{bmatrix} 0 & 0 & e_{31} \\ 0 & 0 & e_{32} \\ 0 & 0 & 0 \\ 0 & e_{24} & 0 \\ e_{15} & 0 & 0 \end{bmatrix} \begin{Bmatrix} E_1 \\ E_2 \\ E_3 \end{Bmatrix} \\ &\quad - \begin{bmatrix} 0 & 0 & q_{31} \\ 0 & 0 & q_{32} \\ 0 & 0 & 0 \\ 0 & q_{24} & 0 \\ q_{15} & 0 & 0 \end{bmatrix} \begin{Bmatrix} H_1 \\ H_2 \\ H_3 \end{Bmatrix} - \begin{Bmatrix} \lambda_1 \\ \lambda_2 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \theta, \end{aligned} \quad (A1)$$

$$\begin{aligned} \begin{Bmatrix} D_1 \\ D_2 \\ D_3 \end{Bmatrix} &= \begin{bmatrix} 0 & 0 & 0 & 0 & e_{15} \\ 0 & 0 & 0 & e_{24} & 0 \\ e_{31} & e_{32} & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{Bmatrix} + \begin{bmatrix} \chi_{11} & 0 & 0 \\ 0 & \chi_{22} & 0 \\ 0 & 0 & \chi_{33} \end{bmatrix} \begin{Bmatrix} E_1 \\ E_2 \\ E_3 \end{Bmatrix} \\ &\quad + \begin{bmatrix} \kappa_{11} & 0 & 0 \\ 0 & \kappa_{22} & 0 \\ 0 & 0 & \kappa_{33} \end{bmatrix} \begin{Bmatrix} H_1 \\ H_2 \\ H_3 \end{Bmatrix} + \begin{Bmatrix} p_1 \\ p_2 \\ p_3 \end{Bmatrix} \theta, \end{aligned} \quad (A2)$$

$$\begin{aligned} \begin{Bmatrix} B_1 \\ B_2 \\ B_3 \end{Bmatrix} &= \begin{bmatrix} 0 & 0 & 0 & 0 & q_{15} \\ 0 & 0 & 0 & q_{24} & 0 \\ q_{31} & q_{32} & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{Bmatrix} + \begin{bmatrix} \kappa_{11} & 0 & 0 \\ 0 & \kappa_{22} & 0 \\ 0 & 0 & \kappa_{33} \end{bmatrix} \begin{Bmatrix} E_1 \\ E_2 \\ E_3 \end{Bmatrix} \\ &\quad + \begin{bmatrix} \gamma_{11} & 0 & 0 \\ 0 & \gamma_{22} & 0 \\ 0 & 0 & \gamma_{33} \end{bmatrix} \begin{Bmatrix} H_1 \\ H_2 \\ H_3 \end{Bmatrix} + \begin{Bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{Bmatrix} \theta, \end{aligned} \quad (A3)$$

$$S = \{\lambda_1 \ \lambda_2 \ 0 \ 0 \ 0\} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{Bmatrix} + \{p_1 \ p_2 \ p_3\} \begin{Bmatrix} E_1 \\ E_2 \\ E_3 \end{Bmatrix} + \{\tau_1 \ \tau_2 \ \tau_3\} \begin{Bmatrix} H_1 \\ H_2 \\ H_3 \end{Bmatrix} + c\theta. \quad (A4)$$

The detailed derivation process for δT , δW_{int} and δW_{ext} is as follows

$$\begin{aligned}\delta T &= - \int_V \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} dV = - \int_V \rho \delta (\mathbf{Z}_u \mathbf{v})^T (\mathbf{Z}_u \ddot{\mathbf{v}}) dV = - \int_V \rho \delta \mathbf{v}^T \mathbf{Z}_u^T \mathbf{Z}_u \ddot{\mathbf{v}} dV \\ &= - \int_V \rho \delta (\mathbf{N} \mathbf{q})^T \mathbf{Z}_u^T \mathbf{Z}_u (\mathbf{N} \ddot{\mathbf{q}}) dV = - \delta \mathbf{q}^T \int_V \rho \mathbf{N}^T \mathbf{Z}_u^T \mathbf{Z}_u \mathbf{N} dV \ddot{\mathbf{q}} \\ &= - \delta \mathbf{q}^T \mathbf{M}_{uu} \ddot{\mathbf{q}},\end{aligned}\quad (\text{A5})$$

$$\begin{aligned}\delta W_{\text{int}} &= \int_V (\delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} - \delta \mathbf{E}^T \mathbf{D} - \delta \mathbf{H}^T \mathbf{B} - \delta \boldsymbol{\theta}^T \mathbf{S}) dV \\ &= \int_V (\delta (\mathbf{B}_u \mathbf{q})^T (\mathbf{C} \boldsymbol{\varepsilon} - \mathbf{e}^T \mathbf{E} - \mathbf{q}^T \mathbf{H} - \boldsymbol{\lambda}^T \boldsymbol{\theta}) - \delta (\mathbf{B}_\phi \boldsymbol{\phi})^T (\mathbf{e} \boldsymbol{\varepsilon} + \chi \mathbf{E} + \kappa \mathbf{H} + \mathbf{p}^T \boldsymbol{\theta}) \\ &\quad - \delta (\mathbf{B}_\psi \boldsymbol{\psi})^T (\mathbf{q} \boldsymbol{\varepsilon} + \kappa \mathbf{E} + \gamma \mathbf{H} + \boldsymbol{\tau}^T \boldsymbol{\theta}) - \delta (\mathbf{B}_T \boldsymbol{\theta})^T (\boldsymbol{\lambda} \boldsymbol{\varepsilon} + \mathbf{p} \mathbf{E} + \boldsymbol{\tau} \mathbf{H} + \mathbf{c} \boldsymbol{\theta})) dV \\ &= \int_V (\delta (\mathbf{B}_u \mathbf{q})^T (\mathbf{C} \mathbf{B}_u \mathbf{q} - \mathbf{e}^T \mathbf{B}_\phi \boldsymbol{\phi} - \mathbf{q}^T \mathbf{B}_\psi \boldsymbol{\psi} - \boldsymbol{\lambda}^T \mathbf{B}_T \boldsymbol{\theta}) \\ &\quad - \delta (\mathbf{B}_\phi \boldsymbol{\phi})^T (\mathbf{e} \mathbf{B}_u \mathbf{q} + \chi \mathbf{B}_\phi \boldsymbol{\phi} + \kappa \mathbf{B}_\psi \boldsymbol{\psi} + \mathbf{p}^T \mathbf{B}_T \boldsymbol{\theta}) \\ &\quad - \delta (\mathbf{B}_\psi \boldsymbol{\psi})^T (\mathbf{q} \mathbf{B}_u \mathbf{q} + \kappa \mathbf{B}_\phi \boldsymbol{\phi} + \gamma \mathbf{B}_\psi \boldsymbol{\psi} + \boldsymbol{\tau}^T \mathbf{B}_T \boldsymbol{\theta}) \\ &\quad - \delta (\mathbf{B}_T \boldsymbol{\theta})^T (\boldsymbol{\lambda} \mathbf{B}_u \mathbf{q} + \mathbf{p} \mathbf{B}_\phi \boldsymbol{\phi} + \boldsymbol{\tau} \mathbf{B}_\psi \boldsymbol{\psi} + \mathbf{c} \mathbf{B}_T \boldsymbol{\theta})) dV \\ &= \delta \mathbf{q}^T (\mathbf{K}_{uu} \mathbf{q} + \mathbf{K}_{u\phi} \boldsymbol{\phi} + \mathbf{K}_{u\psi} \boldsymbol{\psi} + \mathbf{K}_{uT} \boldsymbol{\theta}) + \delta \boldsymbol{\phi}^T (\mathbf{K}_{\phi u} \mathbf{q} + \mathbf{K}_{\phi\phi} \boldsymbol{\phi} + \mathbf{K}_{\phi\psi} \boldsymbol{\psi} + \mathbf{K}_{\phi T} \boldsymbol{\theta}) \\ &\quad + \delta \boldsymbol{\psi}^T (\mathbf{K}_{\psi u} \mathbf{q} + \mathbf{K}_{\psi\phi} \boldsymbol{\phi} + \mathbf{K}_{\psi\psi} \boldsymbol{\psi} + \mathbf{K}_{\psi T} \boldsymbol{\theta}) + \delta \boldsymbol{\theta}^T (\mathbf{K}_{Tu} \mathbf{q} + \mathbf{K}_{T\phi} \boldsymbol{\phi} + \mathbf{K}_{T\psi} \boldsymbol{\psi} + \mathbf{K}_{TT} \boldsymbol{\theta}),\end{aligned}\quad (\text{A6})$$

$$\begin{aligned}\delta W_{\text{ext}} &= \int_V \delta \mathbf{u}^T \mathbf{f}_b dV + \int_\Omega \delta \mathbf{u}^T \mathbf{f}_s d\Omega + \delta \mathbf{u}^T \mathbf{f}_c \\ &\quad - \int_\Omega \delta \boldsymbol{\phi}^T \boldsymbol{\vartheta} d\Omega - \delta \boldsymbol{\phi}^T \mathbf{Q}_c - \int_\Omega \delta \boldsymbol{\psi}^T \mathbf{I} d\Omega - \delta \boldsymbol{\psi}^T \mathbf{I}_c - \int_\Omega \delta \boldsymbol{\theta}^T \bar{\mathbf{Q}} d\Omega \\ &= \int_V \delta (\mathbf{Z}_u \mathbf{N} \mathbf{q})^T \mathbf{f}_b dV + \int_\Omega \delta (\mathbf{Z}_u \mathbf{N} \mathbf{q})^T \mathbf{f}_s d\Omega + \delta (\mathbf{Z}_u \mathbf{N} \mathbf{q})^T \mathbf{f}_c \\ &\quad - \int_\Omega \delta \boldsymbol{\phi}^T \boldsymbol{\vartheta} d\Omega - \delta \boldsymbol{\phi}^T \mathbf{Q}_c - \int_\Omega \delta \boldsymbol{\psi}^T \mathbf{I} d\Omega - \delta \boldsymbol{\psi}^T \mathbf{I}_c - \int_\Omega \delta \boldsymbol{\theta}^T \bar{\mathbf{Q}} d\Omega \\ &= \delta \mathbf{q}^T (\int_V \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_b dV + \int_\Omega \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_s d\Omega + \mathbf{N}^T \mathbf{Z}_u^T \mathbf{f}_c) \\ &\quad + \delta \boldsymbol{\phi}^T (- \int_\Omega \boldsymbol{\vartheta} d\Omega - \mathbf{Q}_c) + \delta \boldsymbol{\psi}^T (- \int_\Omega \mathbf{I} d\Omega - \mathbf{I}_c) + \delta \boldsymbol{\theta}^T (- \int_\Omega \bar{\mathbf{Q}} d\Omega) \\ &= \delta \mathbf{q}^T \mathbf{F}_{ue} + \delta \boldsymbol{\phi}^T \mathbf{F}_{\phi e} + \delta \boldsymbol{\psi}^T \mathbf{F}_{\psi e} + \delta \boldsymbol{\theta}^T \mathbf{F}_{Te}.\end{aligned}\quad (\text{A7})$$

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